



A new adaptive blind background calibration of gain and timing mismatch for a two-channel time-interleaved ADC

Mohammad Issvand Nassiri^a, Mahdi Rezvanyvardom^{b,*}

^a National Iranian South Oil Company (NISOC), Iran

^b Department of Electrical Engineering, Faculty of Engineering, Arak University, Arak, Iran

ARTICLE INFO

Keywords:

Analog-to-Digital Converter
Blind calibration
Gain mismatch
Time-interleaved
Timing mismatch

ABSTRACT

This study investigates a new adaptive blind calibration structure for the calibration of gain and timing mismatch error, which is applied for a two-channel Time-Interleaved Analog-to-Digital Converter (TI-ADC). This technique calibrates the mismatch error in the output of the converter without any interrupt at the typical performance of the TI-ADC. The output of the time-interleaved ADC is approximated by modeling based on Taylor's series. The effect of gain and timing mismatch in the output is represented by the first order of Taylor's series. By using a Least-Mean Square (LMS) and correlation-based algorithm, the approximate coefficients of the Taylor's series are identified. Input signal and its chopped image or input signal and its chopped and delayed image will be correlated in the proposed algorithm. This algorithm can identify the coefficients blindly. It is simpler than the previous similar techniques. Also, this technique does not have the drawbacks of a technique which uses the Taylor's series approximation and filtered-X LMS algorithm. Simulation results confirm these claims and show that by entering six sinusoidal inputs in the whole bandwidth of the proposed technique, 48.2 dB improvement in Spur Free Dynamic Range (SFDR) is achieved.

1. Introduction

Analog-to-Digital Converters (ADCs) are being widely used for signal processing in wide variety of industrial applications such as modern signal processing applications [1]. Since these applications need high-speed ADCs, the time-interleaved architecture has been introduced [1]. A Time-Interleaved ADC (TI-ADC), with sampling rate of F_s , parallelizes M-channel ADCs with sampling rate of $\frac{F_s}{M}$ [2]. A model of two-channel ADC is shown in Fig. 1. If the gain of each channel does not equal with each other and also the time sampling of each channel does not equal exactly $2T_s$ after previous channel, the gain and timing mismatch errors are generated in the output [3–6]. Several techniques have been proposed to eliminate the effect of gain and timing mismatch in the output of TI-ADC such as randomization, two-rank sample and hold and calibration techniques [7–12]. In foreground calibrations, the normal operation of conversion of the TI-ADC has to be interrupted to calibrate the mismatch errors [9]. To avoid the foreground interruptions, several background calibration techniques have been proposed [7, 13–15], which measure mismatch errors by itself and subtract the measured error from the output during the normal operation mode

[16–19]. Recently, calibration technique is the most popular technique for removing these errors. The calibration process categorizes as non-blind and blind methods. The non-blind calibration method needs information about the input signal while the blind calibration method does not need any information about the input signal but applies some limitation on its bandwidth [17].

In this paper, a new blind background calibration technique is presented which calibrates the gain and timing mismatch errors adaptively in a two-channel time-interleaved ADC. In this technique, the gain and timing mismatch errors are modeled using the first-order Taylor's series approximation. The modeled error has two coefficients for the gain and timing mismatch errors. The coefficients are estimated adaptively by correlation between the input tones and the error tones. Finally, the estimated error is subtracted from the output of the two-channel TI-ADC. In this technique, the mismatch errors of the TI-ADC is calibrated blindly while the TI-ADC is at its normal operation mode.

The remainder of this paper is organized as follows. In section 2, gain and timing mismatch errors in a two-channel TI-ADC are discussed. In section 3 a model of two-channel TI-ADC is presented. In section 4, first, drawbacks of calibration by Filtered-X Least Mean Square (FxLMS) and then the proposed calibration technique are presented. Section 4 also

* Corresponding author.

E-mail addresses: issvand.m@nisoc.ir (M.I. Nassiri), m-rezvanyvardom@araku.ac.ir (M. Rezvanyvardom).

<https://doi.org/10.1016/j.mejo.2018.05.008>

Received 12 January 2018; Received in revised form 14 April 2018; Accepted 14 May 2018

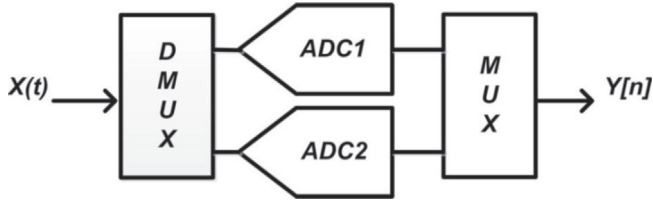


Fig. 1. A model of a two-channel timed-interleaved ADC.

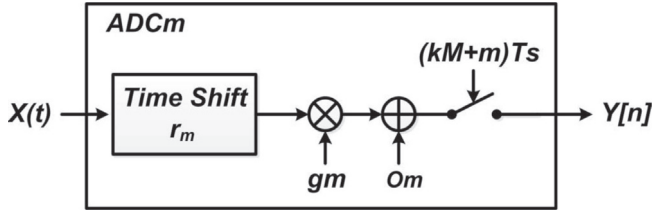


Fig. 2. Mathematical model of a channel in ADC [6].

gives a description of the details of coefficient identification of the proposed calibration technique. The comparison and simulation results are presented in section 5 and 6, respectively and finally section 7 concludes the paper.

2. Mismatch error in a two-channel TI-ADC

According to Fig. 1, when a sinusoidal input signal is used to the system, the output signal in frequency domain is calculated as [26].

$$Y(j\Omega) = \frac{1}{2T_s} \sum_{i=0}^1 \sum_{m=-\infty}^{\infty} \left[g_i \frac{A\pi}{j} \left(\delta \left(\Omega - m \frac{\Omega_s}{2} - \Omega_0 \right) - \delta \left(\Omega - m \frac{\Omega_s}{2} + \Omega_0 \right) \right) e^{-j\Omega_0 \Delta t_i} + o_i 2\pi \delta \left(\Omega - m \frac{\Omega_s}{2} \right) \right] e^{-jm \frac{\Omega_s}{2} T_s} \quad (1)$$

where Δt_i and g_i are the timing errors of ADC_i and the gain of the channel i respectively. Also the value o_i is the offset signal of the channel i . Ω_0 and Ω_s are the input frequency and sampling frequency of the two-channel TI-ADC correspondingly when the input is sinusoidal. Assume that the gains of each channel are different and the other characteristics are identical [25]. If a sinusoidal input signal is applied to the system, the largest difference in channel outputs occurs at the peaks of the sine wave. Timing mismatch causes error in the ADC system, and the largest error occurs in the time domain when the input signal has the largest slew rate, or crosses zero [25].

The mathematical model of a channel of an ADC is shown in Fig. 2. The value $r_m T_s$ shows the timing errors of ADC_m. Also, m is the number of the TI-ADC channel. The input signal is multiplied by g_m which is the gain of the channel. Also the value o_m is the offset of the channel that is added to the input signal. Finally, it is sampled with a sample period of MT_s and constant time shift of mT_s . In this paper offset mismatch error is not assumed. M is the total number of the TI-ADC channels. If in a two-channel TI-ADC the gain and timing errors of the first channel do not equal to the gain and timing errors of second channel, gain and timing mismatch errors are generated [4]. In other words, if $g_0 \neq g_1$ and $r_0 \neq r_1$ the gain and timing mismatch errors are produced. The generated errors in a two-channel TI-ADC produce spurious tones in frequency domain of the output [4], i.e.,

$$\Omega_{error} = k \frac{\Omega_s}{M} \pm \Omega_0 \quad k \in \mathbb{Z}, \quad k \bmod M \neq 0 \quad (2)$$

In equation (2), parameter Ω_{error} is the mismatch error tone frequency. Also Ω_s , Ω_0 are the sampling frequency and the input frequency of the TI-ADC, respectively.

3. Time-interleaved ADC modeling

There are several techniques for modeling the TI-ADC [19,20]. In this section, a model of a two-channel TI-ADC is investigated which is suffering from the gain and timing mismatch errors [19]. In this model, the output of the two-channel TI-ADC is approximated by Taylor's series and is separated by two parts, the first part is an ideal TI-ADC and the second part is a filter that produces the mismatch error in the whole TI-ADC.

The filter approximates the error of the TI-ADC which is an additional signal component in the output frequency spectrum. The additional signal component is an image of the input signal that its amplitude and frequency have a relationship with the input signal [4]. equation (2) refers to the frequency relationship between the input signal and the additional signal component. The ideal TI-ADC is like an ideal ADC but with a difference in performance. The sampling frequency of the ideal TI-ADC is M times faster than an ADC. The ideal TI-ADC produces desired signal $\bar{x}[n]$ and the modeled filter produces the mismatch error signal $e[n]$. The overall output of the TI-ADC is the sum of the modeled-filter output and the ideal TI-ADC output. So for retrieving the desired signal $\bar{x}[n]$, the overall output of the two-channel TI-ADC should be shown as [19]:

$$y[n] = \bar{x}[n] + e[n] \quad (3)$$

The Discrete-Time Fourier Transform (DTFT) of the output signal of a two-channel TI-ADC $y[n]$ is given by Ref. [19]:

$$Y(e^{j\omega}) = \sum_{i=0}^1 \check{V}_i(e^{j(\omega-k\pi)}) X(e^{j(\omega-k\pi)}) \quad (4)$$

where

$$\check{V}_i(e^{j\omega}) = \frac{1}{2} \sum_{q=0}^1 V_q(e^{j\omega}) e^{-jk\pi q} \quad (5)$$

and $V_q(e^{j\omega})$ is the 2π -periodic extension of $\hat{V}_q(j\Omega)$, which is the analog channel frequency response. By taking DTFT equation (3) can be rewritten as:

$$Y(e^{j\omega}) = \bar{X}(e^{j\omega}) + E(e^{j\omega}) \quad (6)$$

which $\bar{X}(e^{j\omega})$, $E(e^{j\omega})$ are the desired output signal and the error signal, respectively.

$$\bar{X}(e^{j\omega}) = \check{V}_0(e^{j\omega}) X(e^{j\omega}) \quad (7)$$

where $X(e^{j\omega})$ and $\check{V}_0(e^{j\omega})$ are the input signal and the average channel frequency response, respectively. The error signal in (6) can be expressed as follows:

$$E(e^{j\omega}) = \check{S}(e^{j(\omega-\pi)}) \bar{X}(e^{j(\omega-\pi)}) \quad (8)$$

where $\check{S}(e^{j\omega})$ is the normalized frequency response mismatch. $\check{S}(e^{j\omega})$ is

calculated as:

$$\check{S}(e^{j\omega}) = \frac{\check{V}_1(e^{j\omega})}{\check{V}_0(e^{j\omega})} \quad (9)$$

By taking the inverse DTFT of (6)–(8) the time-domain input/output relationships of a two-channel TI-ADC are obtained as the following.

$$\bar{x}[n] = \check{y}_0[n] * x[n], \quad e[n] = \check{s}[n] * \bar{x}[n](-1)^n \quad (10)$$

As represented by (10) to get the desired output signal $\bar{x}[n]$, the error signal $e[n]$ should be subtracted from the output signal $y[n]$. The error signal $e[n]$ is resulted by the chopped desired output signal $\bar{x}[n](-1)^n$ convolved by the discrete-time filter $\check{s}[n]$. For modeling the frequency response mismatch, the discrete-time frequency response $\check{S}(e^{j\omega})$ can be represented as a Pth order polynomial [19], i.e.,

$$\check{S}(e^{j\omega}) = \sum_{r=0}^P \check{C}_r D_r(e^{j\omega}) = \sum_{r=0}^P \check{C}_r(j\omega)^r \quad -\pi < \omega < \pi \quad (11)$$

where $\check{C}_r$ is the r th order coefficient of polynomial series and $D_r(e^{j\omega})$ is the discrete-time representation of a band limited r th order continuous-time differentiator. Since, the gain and timing mismatch errors are the most important and the most effective errors in the frequency response mismatch of the TI-ADC, just the gain and timing mismatch errors are considered. In the time-domain, the gain mismatch and the timing mismatch errors are according to the scale of the desired signal and the derivative of the desired signal, respectively. Therefore, the value of r is chosen to be unity in (11). Thereby (11) can be rewritten as

$$\check{S}(e^{j\omega}) = \check{C}_0 D_0(e^{j\omega}) + \check{C}_1 D_1(e^{j\omega}) \quad (12)$$

where according to (11), $D_0(e^{j\omega})$ is equal to 1 and $D_1(e^{j\omega})$ is equal to $j\omega$, also in (12) $\check{C}_0$ is the gain mismatch coefficient, which renames as $\check{C}_g$, and $\check{C}_1$ is the timing mismatch coefficient which renames as $\check{C}_t$ and By taking inverse DTFT of it the discrete-time filter $\check{S}(e^{j\omega})$ is obtained as

$$\check{s}[n] = \check{C}_g \delta[n] + \check{C}_t d[n] \quad (13)$$

where $d[n]$ is the inverse DTFT of $j\omega$ and an impulse response of the derivative filter, i.e.,

$$d[n] = \begin{cases} \frac{\cos(n\pi)}{n} & n \neq 0 \\ 0 & n = 0 \end{cases} \quad (14)$$

The coefficients of the derivative filter $d[n]$ are determined by multiplying the exact coefficients by a window function to reduce the influence of reduction error. Substituting (16) in (11) gives

$$e[n] = \check{C}_g \bar{x}[n](-1)^n + \check{C}_t d[n] * \bar{x}[n](-1)^n \quad (15)$$

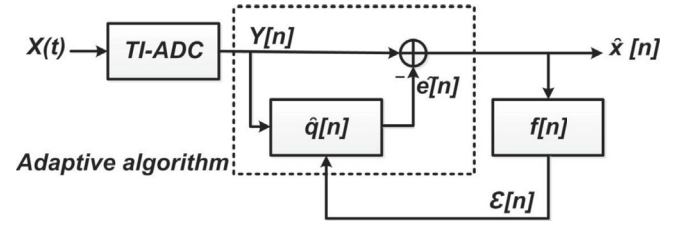


Fig. 3. Calibration by Fx-LMS algorithm [22].

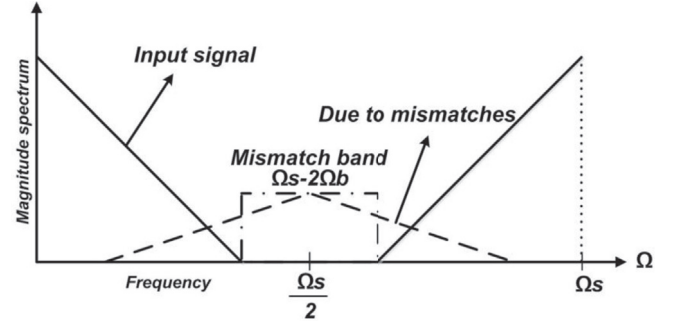


Fig. 4. Magnitude spectrum of the output of a two-channel TI-ADC for calibration by FxLMS algorithm [22].

the input signal and its delayed image, those drawbacks are eliminated. To compensate the output of TI-ADC suffering from the gain and the timing mismatch errors, the error signal $e[n]$ should be subtracted from the output signal. Therefore, the approximation of error signal $\hat{e}[n]$ should be resulted. For this reason, the approximation of error signal $\hat{e}[n]$ is approximated by estimation of the gain and the timing coefficients $\hat{C}_g[n]$, $\hat{C}_t[n]$, respectively.

The reconstructed signal $\hat{x}[n]$ is resulted by subtraction the estimated approximation error $\hat{e}[n]$ from the output of TI-ADC suffering from gain and timing mismatch errors $y[n]$. Therefore, the error $\hat{e}[n]$ is given by

$$\hat{e}[n] = \hat{y}_g[n] \hat{C}_g[n] + \hat{y}_t[n] \hat{C}_t[n] = \hat{e}_g[n] + \hat{e}_t[n] \quad (16)$$

where the signal $\hat{y}_g[n]$ and $\hat{y}_t[n]$ are chopped signal and filtered chopped signal, respectively. Thus:

$$\hat{y}_g[n] = y[n](-1)^n \quad (17)$$

$$\hat{y}_t[n] = (y[n] * d[n])(-1)^n \quad (18)$$

As mentioned in section 3, the output signal $y[n]$ is separated into a desired output signal $\bar{x}[n]$ and an error signal $e[n]$. By substituting (10), (16)–(18) and (15) gives:

$$\hat{x}[n] = \bar{x}[n] + \bar{x}[n] \left((-1)^n \hat{C}_g[n] + (\bar{x}[n] * d[n]) \right) (-1)^n \hat{C}_t[n] - (\bar{x}[n] (-1)^n \hat{C}_g[n] - (e[n] (-1)^n \hat{C}_g[n] - (\bar{x}[n] * d[n]) (-1)^n \hat{C}_t[n] - (e[n] * d[n]) (-1)^n \hat{C}_t[n]) \quad (19)$$

4. Adaptive blind calibration

This section shows estimation of the coefficients of the error model approximation $\hat{C}_g[n]$, $\hat{C}_t[n]$ by FxLMS algorithm. Also drawbacks of the estimation by FxLMS algorithm are shown, then by using a technique based on correlation between the input signal and its image and between

Therefore, if the estimated coefficient of gain and timing errors $\hat{C}_g[n]$, $\hat{C}_t[n]$ equal to the actual coefficients of gain and timing errors $\check{C}_g[n]$, $\check{C}_t[n]$ respectively, i.e., $\check{C}_g[n] = \hat{C}_g[n]$ and $\check{C}_t[n] = \hat{C}_t[n]$, the reconstructed output signal is calculated as:

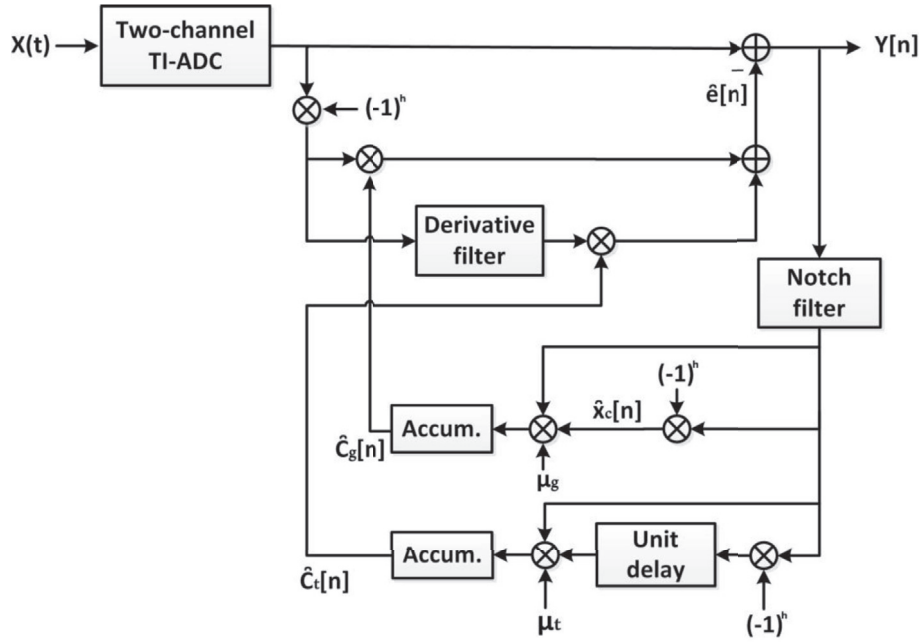
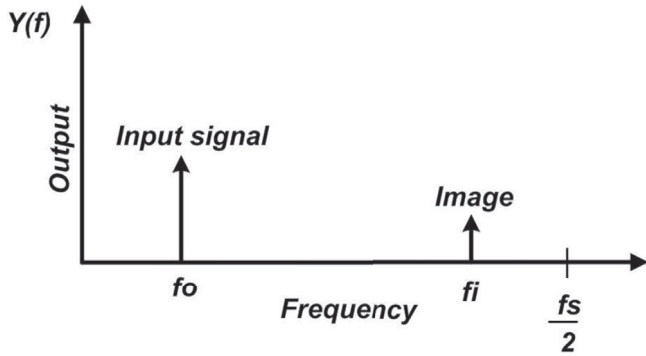
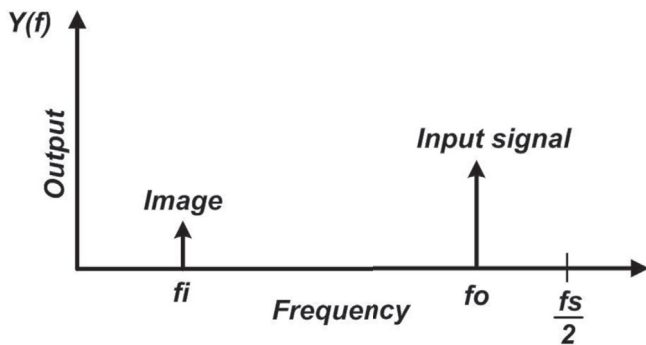


Fig. 5. Calibration by the proposed algorithm.



(a)



(b)

Fig. 6. Output spectrum for (a) a two-channel TI-ADC and (b) chopped output spectrum of it [23].

$$\hat{x}[n] = \bar{x}[n] - (e[n](-1)^n \hat{C}_g[n] - (e[n] * d[n])(-1)^n \hat{C}_t[n]) \quad (20)$$

whatever except $\bar{x}[n]$ in (20) are the remaining error signal after calibration. As can be seen in Ref. [21] the energy of the remaining error signal is much smaller than the energy of the mismatch error signal $e[n]$, so the reconstructed signal $\hat{x}[n]$ is a good approximation of the desired output signal $\bar{x}[n]$.

4.1. Coefficients estimation by FxLMS algorithm

Fig. 3 shows the basic concept of calibration by FxLMS algorithm. The coefficients estimation is obtained by Ref. [19]:

$$\hat{C}[n+1] = \hat{C}[n] + \mu \cdot e[n] \cdot y_d[n-k] \quad (21)$$

where k is the delay of the linear-phase high-pass filter $f[n]$ and μ is the step size of LMS algorithm. As shown in Fig. 3, for identifying the coefficients of the error model approximation $\hat{s}[n]$, the output signal $y[n]$ is passed through a high-pass filter $f[n]$. In order to result in filtered-output signal $\varepsilon[n]$, it is entered an adaptive algorithm which is based on a LMS algorithm. Since the filtered-output signal $\varepsilon[n]$ is the input of the LMS block, the algorithm is called FxLMS [19].

For a correct estimation, the filtered-output signal $\varepsilon[n]$ should have only the error tones because only correlation between the error tones in $\varepsilon[n]$ and the input tones in $y[n]$ results correct coefficients of the error model approximation $\hat{s}[n]$ [19]. To get only the error tones in $\varepsilon[n]$, the bandwidth of input signal should have maximum frequency Ω_b which is a frequency less than $\Omega_s/2$. As a result, an area in frequencies Ω_b to $\Omega_s/2$ is formed that has only the mismatch error tones. This area is called the mismatch band. The magnitude frequency spectrum of the output of a two-channel TI-ADC is shown in Fig. 4 in order to be understand more. Since the maximum frequency of the input Ω_b is less than niquist rate $\Omega_s/2$, thereby some of the input bandwidth would be redundant. So the first drawback of calibration by the FxLMS algorithm is a limitation on the input bandwidth and wastage of it.

Refer to (2), in a two-channel TI-ADC, the error tones in the mismatch band are resulted from the input signal with frequencies from 0 to $(\Omega_s/2 - \Omega_b)$. So if in a two-channel TI-ADC the input signal does not have any frequency component of 0 to $(\Omega_s/2 - \Omega_b)$, then no error tone appears in

the mismatch band. Thus, the filtered-output signal $\varepsilon[n]$ will become zero, so in this case, the FxLMS algorithm based on (21) cannot identify the mismatch error.

Therefore, the second drawback of calibration by the FxLMS algorithm is the inability to calibrate the mismatch error when no input tones are not in frequencies of 0 to $(\Omega_s/2 - \Omega_b)$.

4.2. Coefficients identification by proposed technique

To get the reconstructed signal $\hat{x}[n]$, the coefficients of gain and timing errors $\hat{C}_g[n]$, $\hat{C}_t[n]$ should be estimated. To do this without any drawbacks of the calibration by the FxLMS in Ref. [19] as mentioned

$$\hat{e}[n] = \hat{e}_g[n] = y_g[n] \hat{C}_g[n] \quad (28)$$

By substituting (27) in (17) and then the result in (28), we can rewrite (28) as

$$\hat{e}[n] = G_1 \hat{C}_g V[n] \delta[n \bmod 2] - G_2 \hat{C}_g V[n] \delta[(n \bmod 2) - 1] \quad (29)$$

Thus

$$\begin{aligned} \hat{x}[n] = & G_1 V[n] \delta[n \bmod 2] + G_2 V[n] \delta[(n \bmod 2) - 1] \\ & - G_1 \hat{C}_g V[n] \delta[n \bmod 2] + G_2 \hat{C}_g V[n] \delta[(n \bmod 2) - 1] \end{aligned} \quad (30)$$

and as shown in Fig. 5 $\hat{x}_c[n]$ is

$$\hat{x}_c[n] = \hat{x}[n] (-1)^n = G_1 V[n] \delta[n \bmod 2] - G_2 V[n] \delta[(n \bmod 2) - 1] - G_1 \hat{C}_g V[n] \delta[n \bmod 2] + G_2 \hat{C}_g V[n] \delta[(n \bmod 2) - 1] \quad (31)$$

earlier, a technique based on correlation between input signal and chopped signal/filtered chopped signal is used in the proposed technique [21]. This technique is substituted for the FxLMS algorithm. Fig. 5 shows the technique that is used in this paper. As shown in Fig. 5 for identifying the coefficient of the gain error $\hat{C}_g$, the reconstructed signal $\hat{x}[n]$ should be multiplied by $(-1)^n$. By doing this, the reconstructed signal $\hat{x}[n]$ shifts by $F_s/2$ in frequency domain. Fig. 6(a) shows an output signal of a two-channel TI-ADC in frequency domain. By shifting the reconstructed $\hat{x}[n]$ (the final reconstructed signal has not resulted yet) frequency f_i shifts to f_o and f_o shifts to f_i . Then the chopped signal $\hat{x}_c[n]$ in Fig. 6(b) is resulted. Subsequently, the reconstructed signal $\hat{x}[n]$ and its chopped signal $\hat{x}_c[n]$ are multiplied in the time domain that results a DC component with respect to the gain mismatch.

By the proposed technique, when an input just with frequency at $F_s/4$ enters to the identification block, a problem occurs [21]. In this case, the result of multiplying $\hat{x}[n]$ and $\hat{x}_c[n]$ may produces a dc component even though there is not any mismatch error in the TI-ADC that the result of coefficient identification will be spoiled. So for solving this problem, a notch filter is used to prevent the input signal at frequency $F_s/4$ from entering to the identification block.

The output of a two-channel TI-ADC without assuming notch filter and the timing mismatch can be written as

$$y[n] = u_1[n] + u_2[n] \quad (22)$$

where

$$u_1[n] = G_1 V[n] \delta[n \bmod 2] \quad (23)$$

$$u_2[n] = G_2 V[n] \delta[(n \bmod 2) - 1] \quad (24)$$

where $V[n]$ is the ideal output of the TI-ADC. Also

$$\delta[n \bmod 2] = \begin{cases} 1 & \text{for } n = \text{even} \\ 0 & \text{for } n = \text{odd} \end{cases} \quad (25)$$

$$\delta[(n \bmod 2) - 1] = \begin{cases} 0 & \text{for } n = \text{even} \\ 1 & \text{for } n = \text{odd} \end{cases} \quad (26)$$

which $u_1[n]$ and $u_2[n]$ are the outputs of the first and second channel of TI-ADC, respectively and G_1, G_2 are the gains of them. Substituting (23), (24) in (22) gives

$$y[n] = G_1 V[n] \delta[n \bmod 2] + G_2 V[n] \delta[(n \bmod 2) - 1] \quad (27)$$

The error $\hat{e}[n]$ is equal to

Since for analyzing the gain mismatch error the timing mismatch error is not assumed. Thus, based on (20) the update equation based on Fig. 5 is

$$\hat{C}_g[n+1] = \hat{C}_g[n] + \mu \{ \hat{x}[n] \cdot \hat{x}_c[n] \} \quad (32)$$

If $V = A \cos(\omega_0 t)$, $\omega_0 \neq \omega_s/4$, $\omega_0 \neq 0$;

$$\overline{V^2} = A^2/2 \quad (33)$$

$$\overline{V^2 \delta[n \bmod 2]} = A^2/4 \quad (34)$$

and

$$\overline{V^2 \delta[(n \bmod 2) - 1]} = A^2/4 \quad (35)$$

By taking expectation of (32) and using (33)–(35) it becomes

$$\begin{aligned} \overline{\hat{C}_g[n+1]} = & \overline{\hat{C}_g[n]} + \mu_g \frac{A^2}{4} \left\{ \left(G_1^2 - 2G_1 \overline{\hat{C}_g[n]} + G_1^2 \overline{\hat{C}_g^2[n]} \right) - \left(G_2^2 \right. \right. \\ & \left. \left. - 2G_2 \overline{\hat{C}_g[n]} + G_2^2 \overline{\hat{C}_g^2[n]} \right) \right\} \end{aligned} \quad (36)$$

and by achieving the convergence $\lim_{n \rightarrow \infty} \overline{\hat{C}_g[n+1]} = \lim_{n \rightarrow \infty} \overline{\hat{C}_g[n]}$ and taking

into account the fact that $\hat{C}_g \ll G_1, G_2$ and $\Delta g = G_1 - G_2$ and dissolving it, it will be

$$\lim_{n \rightarrow \infty} \overline{\hat{C}_g[n]} \approx \frac{\Delta g}{2G_1} \quad (37)$$

By multiplying $\Delta g/2G_1$ by gain of each channel G_i , $i = 1, 2$ and taking into account the fact that $\hat{C}_g$ is much less than G_1, G_2 , and also $\frac{G_2}{G_1} \cong 1$ and is much more than $\frac{\Delta g}{2}$ (37) will be

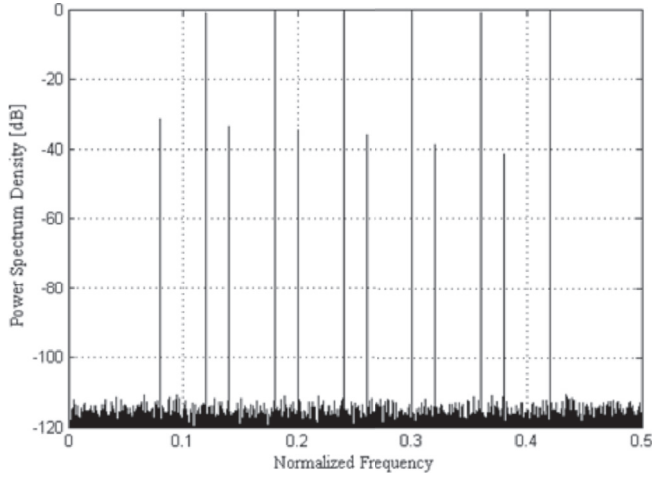
$$\lim_{n \rightarrow \infty} \overline{\hat{C}_g[n]} G_i \approx \frac{\Delta g}{2} \quad (38)$$

When i is equal to unit ($i = 1$), it will be $\frac{\Delta g}{2G_1} G_1$ that it is clear that it is $\frac{\Delta g}{2}$. When i equal to 2 ($i = 2$), it is equal to $\frac{\Delta g}{2G_1} G_2$. The difference between the gain of the channels 1 and 2 ($\Delta g = G_1 - G_2$) is equal to the gain mismatch error which its value is 10% ($\Delta g = 0.1$) in the worst case. Therefore, in this case, the value of $\frac{G_2}{G_1}$ is approximately equal to unit. So in both cases, it is approximately the same $\frac{\Delta g}{2}$. The update equation of the timing coefficient based on Fig. 5 is

Table 1

The performance summary and comparison.

	Ref [19]	Ref [21]	Ref [22]	This paper
Hardware	simple	Not simple	simple	simplest
SNR improvement	13.6 dB	–	–	49 dB
SFDR improvement	23 dB	23.6 dB	30 dB	48 dB
Gain mismatch convergence	10^6 Samples	–	4×10^4 Samples	10^6 Samples
Timing mismatch convergence	10^5 Samples	–	5×10^4 Samples	10^6 Samples

**Fig. 7.** Spectrum of a two-channel TI-ADC output suffering from mismatch error.

$$\hat{C}_t[n+1] = \hat{C}_t[n] + \mu \{ \hat{x}[n] \cdot \hat{x}_c[n-1] \} \quad (39)$$

The identification of the timing and gain error coefficients are the same way, except that for timing error coefficient, after resulting $\hat{x}_c[n]$, it is delayed by one sample. Since the chopped reconstructed signal $\hat{x}_c[n]$ and the reconstructed signal $\hat{x}[n]$, in spite of timing error, are orthogonal (until the final reconstructed signal is resulted), for preventing zero values of the multiplied signal, one sample delay is given to the reconstructed signal $\hat{x}_c[n]$. Since for analyzing the timing mismatch error, the gain mismatch error based on (16) is not assumed and also according to (16) $\hat{x}[n]$ becomes

$$\hat{x}[n] = y[n] - (y[n] * d[n] \cdot (-1)^n) \quad (40)$$

Referring to (31) $\hat{x}_c[n]$ is resulted. Also, by achieving the convergence and by taking account that the output of a two-channel TI-ADC suffering from timing error Δt is obtained as [23]

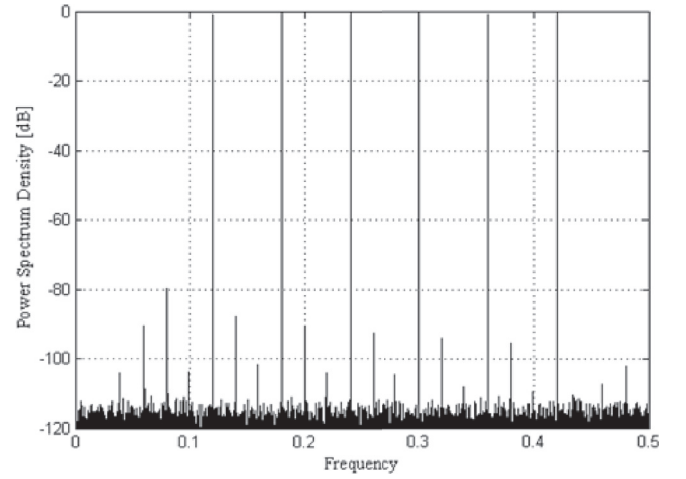
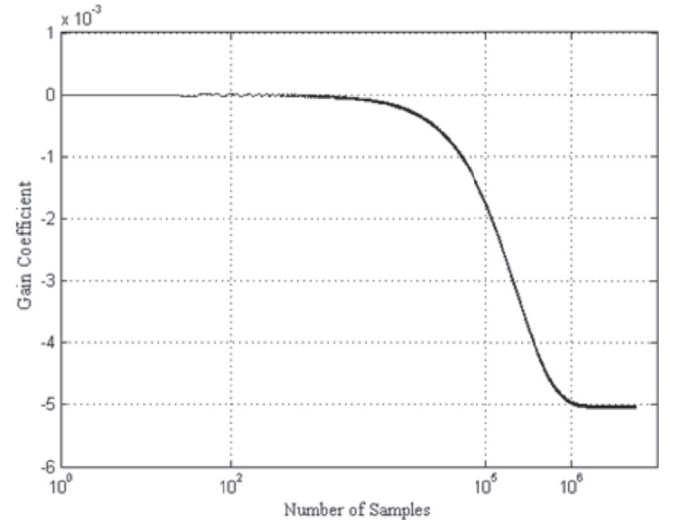
$$y[n] \approx \cos(\omega_0 nT) + \frac{\omega_0 \Delta t}{4} \left\{ \sin\left(\left(\frac{\omega_s}{2} - \omega_0\right)nT\right) + \sin\left(\left(\frac{\omega_s}{2} + \omega_0\right)nT\right) \right\} \quad (41)$$

expectation of the final value of $\hat{C}_t[n+1]$ in (41) will be

$$\lim_{n \rightarrow \infty} \overline{\hat{C}_t[n]} = \frac{\Delta t}{2T_s} \quad (42)$$

5. Comparison

In this section, there is a comparison between the proposed technique in this paper and the three papers [19,21,22] which are almost similar to each other. As mentioned before, calibration process is a two-step process to eliminate the mismatch error in the output. The first step is the identification block that estimates the mismatch error and the second

**Fig. 8.** Spectrum of a two-channel TI-ADC compensated output.**Fig. 9.** Convergence behavior of the estimated gain coefficient.

step is the compensation block that subtracts the estimated error from the output. In case of the gain mismatch error, the identification blocks of the proposed structures in Refs. [19,21,22] and the proposed technique in this paper are almost the same. In Refs. [19] [22], and this paper, the estimated error is applied in the output of the TI-ADC but in Ref. [21] the gain mismatch is corrected by a coefficient in each channel of the TI-ADC.

The calibration method of the timing mismatch error in the proposed structure is the main difference with other calibration methods which are presented in Refs. [19,21,22]. As mentioned in Section 4, the input signal and the image of the input signal, which is produced due to the timing mismatch error, are orthogonal. The identification blocks in these four papers are based on the correlation between the input signal and the mismatch error. Therefore, the orthogonal state should be eliminated for estimating the timing mismatch error. For doing that, a derivative filter is used in Refs. [19] and [22] that has benefit of high convergence. Whereas in Ref. [21] and the proposed technique, a unit delay is used that has a simple implementation structure.

The identification block in the paper [19] has a frequency bandwidth that the error signal just exists there to correlate with the input signal. The drawback of this technique is elaborated before. By comparing the identification block between this paper and the paper [22], it is obvious that the implementation of a derivative filter is harder than a unit delay. Also there are more filters in the paper [22] comparing by this paper but there is not any notch filter to eliminate the frequency $F_s/4$, which makes

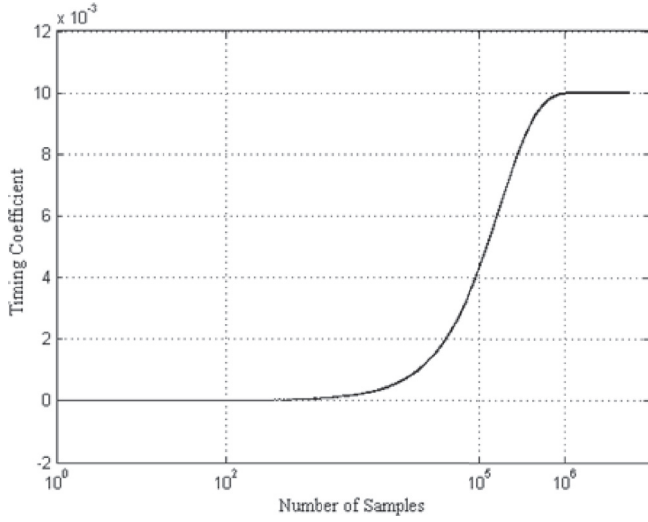


Fig. 10. Convergence behavior of the estimated timing coefficient.

error in the identification block. In the compensation block, a fractional delay is used in Ref. [21] for applying the estimated error. This technique has the most precision but it is more difficult to implement. The compensation blocks that are used in this paper and the paper [22] is as same as the paper [19] except that, the first two terms of the Taylor expansion series are used to approximate the change which is called a second order approximation. In the calibration method presented in Ref. [23], the normal operation of the TI-ADC has to be interrupted while in this work, the calibration process and the normal operation of the TI-ADC works simultaneously. In Ref. [24], the SFDR is improved by 30 dB by spreading out the distortions to the noise floor. It means the distortions is not eliminated but converted to the noise while this work detects the error and eliminates that.

The performance of the proposed circuit is summarized in Table 1 with the previously reported calibration techniques.

6. Simulation results

The proposed calibration technique was simulated by MATLAB for a 16-bit two-channel Time-Interleaved ADC.

A sinusoidal signal with frequencies of $0.12F_s$, $0.1802F_s$, $0.2402F_s$, $0.3003F_s$, $0.3604F_s$ and $0.4205F_s$ is given to a two-channel TI-ADC. F_s is the frequency sampling of two-channel TI-ADC. The first and second gain

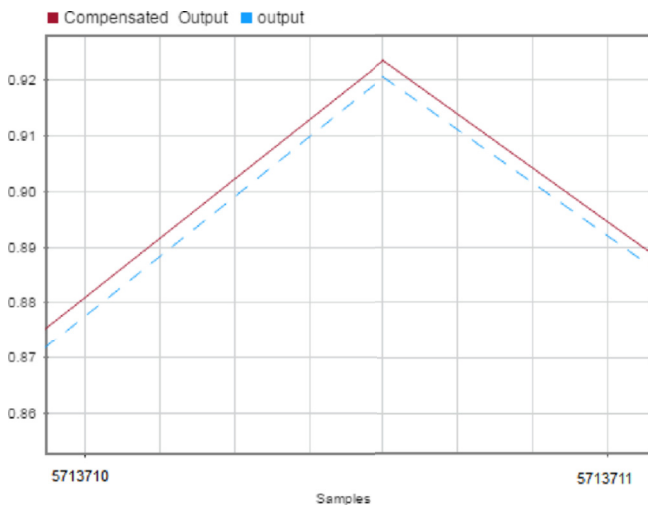


Fig. 11. A part of the compensated output and the output in time domain.

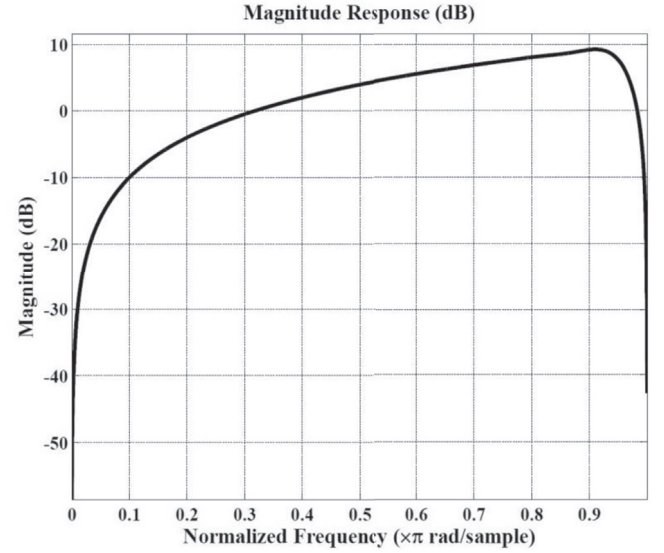


Fig. 12. Frequency response of the derivative filter.

channels of TI-ADC are 1.01 and 1, respectively which means the gain mismatch error is -0.01 . Also, the first and second timing error of channels of TI-ADC are 0 and 0.02, respectively which means the timing mismatch error is 0.02.

Fig. 7 shows the output frequency spectrum of the TI-ADC. As shown in this figure, that is before the calibration, there are 6 spurious tones which the maximum amount of those is -31.5 dB. Signal-to-Noise Ratio(SNR) in this figure is 34.5 dB. In this simulation, a 40th order FIR filter for the derivative filter is used in MATLAB. The coefficients of the filter are determined by multiplying the exact values of an ideal derivative filter $d[n]$, described in section 3, by a hanning window, to reduce a reduction error. The adaptation step size is $\mu_t = \mu_g = 2^{-15}$. Fig. 8 shows the compensated signal. The error components of the output frequency spectrum of the TI-ADC, due to channel mismatch, almost are removed. In Fig. 8, that is after the calibration, the maximum amount of mismatch error tones is -79.6 dB and SNR is 83.4 dB. Fig. 9 and Fig. 10 show the convergence of the gain and timing coefficients of error model as described in Section 4, which they finally converge at 10^6 samples. As shown in Fig. 8, there are 48.1 dB improvements in Spur Free Dynamic Range(SFDR) in the compensated output. Also there is 49 dB improvements in Signal-to-Noise Ratio(SNR).

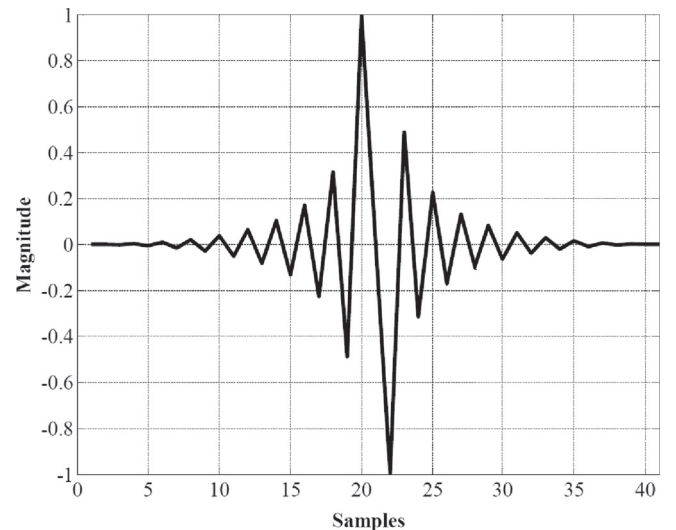


Fig. 13. Time response of the derivative filter.

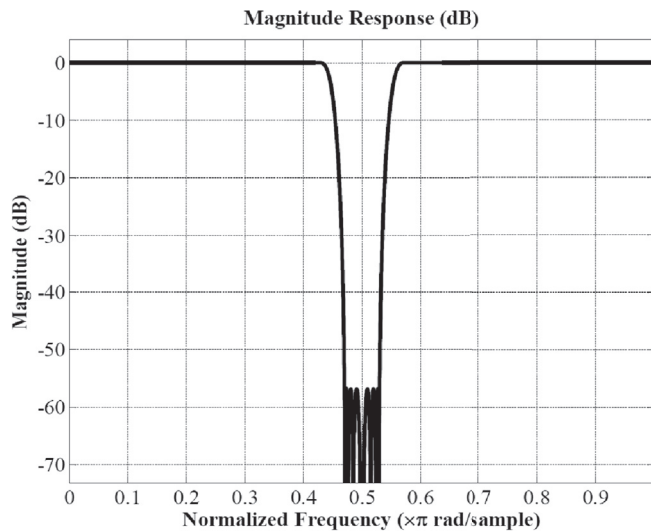


Fig. 14. Frequency response of the notch filter.

For showing that the proposed calibration technique is overcoming the two drawbacks of [19], there are not any component between frequencies of 0 to $0.1F_s$ in the input signal bandwidth. In order to show that the proposed calibration technique has ability to calibrate the error mismatch without any component frequency in 0 to $0.1F_s$.

It means the proposed calibration technique does not have the second drawback of calibration by FxLMS. The sixth sinusoidal signal is in the frequencies from $0.4F_s$ to $0.5F_s$ which is called mismatch band in FxLMS technique. As a result, there is no limitation on the input signal bandwidth for calibration by the proposed technique. So the two drawbacks in Ref. [19] are eliminated by the proposed calibration technique. Fig. 11 shows a part of the compensated output and the output in time domain to perceive how much calibration effect on the output. Fig. 12 shows the frequency response of the derivative filter while Fig. 13 shows the time response of the derivative filter. Also, for the proposed circuit, the frequency response of the notch filter is presented in Fig. 14.

7. Conclusion

An adaptive blind background calibration which is based on Taylor's series approximation [19] is demonstrated in this paper that proved it has two drawbacks. The first drawback is the limitation on the bandwidth of input and the second drawback is the inability to calibrate the error in the whole bandwidth. It means that the calibration in Ref. [19] calibrates the mismatch errors but it cannot calibrate the mismatch errors in some part of the bandwidth of the input signal. The paper explained that those drawbacks are because of coefficients estimation block in that technique. So in this paper a technique for estimation of gain and timing mismatch coefficients replaced with that technique to eliminate those drawbacks. In this paper a new technique for calibration of a two-channel TI-ADC is presented which has less limitation on the input signal. Therein, the proposed technique calibrates the errors without those drawbacks. The part of the calibration, which compensates the error in the calibration, works based on the polynomial-based structures and, the part which estimates the error, works based on the correlation between the input and the error tone. The estimation and the compensation blocks of the calibration are in the output of the TI-ADC so it is a background calibration which operates in the digital domain, so it benefits from digital signals. Since the estimation block has a feedback from the reconstructed output, it calibrates the error adaptively. The technique can adaptively and blindly estimate the error without those drawbacks. Comparing this technique with [21] states that the technique has a simpler compensator filter. The filter in this technique is a derivative filter, while [21] used a fractional delay filter. Also, this technique has less complexity compared

with [22] in time coefficient estimation block. Ref. [22] used a derivative filter while the proposed technique used a unit delay. The simulation results have confirmed that the SFDR in the output is improved with a suitable amount.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.mejo.2018.05.008>.

References

- [1] W. Black, D. Hodges, Time interleaved converter arrays, *IEEE J. Solid State Circ.* 15 (1980) 1022–1029.
- [2] C. Vogel, H. Johansson, Time-interleaved analog-to-digital converters. Status and future directions, in: *Proc. 2006 IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2008, pp. 3386–3389.
- [3] A. Petraglia, S.K. Mitra, Analysis of mismatch effects among A/D converters in a time-interleaved waveform digitizer, *IEEE Trans. Instrum. Measure.* 40 (5) (1991) 831–835.
- [4] N. Kurosawa, H. Kobayashi, K. Maruyama, H. Sugawara, K. Kobayashi, Explicit analysis of channel mismatch effects in time-interleaved ADC systems, *IEEE Trans. Circ. Syst. I* 48 (3) (2001) 261–271.
- [5] J. Elbornsson, F. Gustafsson, J.-E. Eklund, Analysis of mismatch noise in randomly interleaved ADC system, in: *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, vol. 6, 2003, pp. 277–280.
- [6] C. Vogel, The impact of combined channel mismatch effects in time-interleaved ADCs, *IEEE Trans. Instrum. Measure.* 54 (1) (2005) 415–427.
- [7] K.C. Dyer, D. Fu, S.H. Lewis, P.J. Hurst, An analog background calibration technique for time-interleaved analog-to-digital converters, *IEEE J. Solid State Circ.* 33 (12) (1998) 1912–1919.
- [8] P. Zhang, F. Ye, B. Yu, L. Luo, J. Ren, Mixed-signal calibration of sample-time error in time-interleaved ADCs, *Electron. Lett.* 47 (9) (2011) 533–535.
- [9] K. El-Sankary, A. Assi, M. Sawan, New sampling method to improve the SFDR of time-interleaved ADCs, in: *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS 2003)*, vol. 1, 2003, pp. 833–836.
- [10] K. Poulton, J.J. Corcoran, T. Hornak, A 1-GHz 6-bit ADC system, *IEEE J. Solid State Circ.* 22 (6) (1987) 962–970.
- [11] H. Jin, E.K. Lee, A digital-background calibration technique for minimizing timing-error effects in time-interleaved ADCs, *IEEE Trans. Circ. Syst. II Analog Digit. Signal Process.* 47 (7) (2000) 603–613.
- [12] A. Haftbaradaran, K.W. Martin, A background sample-time error calibration technique using random data for wide-band high-resolution time-interleaved ADCs, *IEEE Trans. Circ. Syst. II* 55 (3) (2008) 234–238.
- [13] Chung-Yi Wang, J.-T. Wu, A background timing-skew calibration technique for time-interleaved analog-to-digital converters, *IEEE Trans. Circ. Syst. II* 53 (4) (2006) 299–303.
- [14] S.-H. Lee, B.-S. Song, Digital-domain calibration of multistep analog-to-digital converters, *IEEE J. Solid State Circ.* 27 (12) (1992) 1679–1688.
- [15] A.N. Karanickolas, H.-S. Lee, K. Barracania, A 15-b 1-Msample/s digitally self-calibrated pipeline ADC, *IEEE J. Solid State Circ.* 28 (12) (1993) 1207–1215.
- [16] Seung-Hoon Lee, B.-S. Song, Interstage gain proration technique for digital-domain multi-step ADC calibration, *IEEE Trans. Circ. Syst. II* 41 (1) (1994) 12–18.
- [17] P. Satazadeh, B.C. Levy, P.J. Hurst, Adaptive semibinary calibration of bandwidth mismatch for two-channel time-interleaved ADCs, *IEEE Trans. Circ. Syst. I* 56 (9) (2009) 2075–2088.
- [18] C. Vogel, S. Mendel, A flexible and scalable structure to compensate frequency response mismatches in time-interleaved ADCs, *IEEE Trans. Circ. Syst. I* 56 (11) (2009) 2463–2475.
- [19] S. Saleem, C. Vogel, Adaptive blind background calibration of polynomial-represented frequency response mismatches in a two-channel time-interleaved ADC, *IEEE Trans. Circ. Syst. I* 58 (6) (2011) 1300–1310.
- [20] C. Vogel, G. Kubin, Modeling of time-interleaved ADCs with nonlinear hybrid filter banks, *AEU-Int. J. Electron. Commun.* 59 (2005) 288–296.
- [21] S.M. Jamal, D. Fu, N.-J. Chang, P.J. Hurst, S.H. Lewis, A 10-b 120-Msample/s time-interleaved analog-to-digital converter with digital background calibration, *IEEE J. Solid State Circ.* 37 (12) (2002) 1618–1627.
- [22] Junya Matsuno, et al., All-digital background calibration technique for time-interleaved ADC using pseudo aliasing signal, *IEEE Trans. Circ. Syst. I: Regular Pap.* 60 (2013) 1113–1121.
- [23] Y. Park, J. Kim, C. Kim, A scalable bandwidth mismatch calibration technique for time-interleaved ADCs, *IEEE Trans. Circ. Syst. I: Regular Pap.* 63 (11) (2016) 1889–1897.
- [24] H. Mafi, M. Yargholi, M. Yavari, Digital blind background calibration of imperfections in time-interleaved ADCs, *IEEE Trans. Circ. Syst. I: Regular Pap.* 64 (6) (2017) 1504–1514.
- [25] N. Kurosawa, H. Kobayashi, K. Maruyama, H. Sugawara, K. Kobayashi, Explicit analysis of channel mismatch effects in time-interleaved ADC systems, *IEEE Trans. Circ. Syst. I: Fund. Theor. Appl.* 48 (3) (2001) 261–271.
- [26] R. van Otten, Timing Correction of Time-Interleaved ADCs, Department of Electrical Engineering Mixed-signal Microelectronics group, Eindhoven University of Technology, 2009.